

An explicit affirmative solution to Problem 4 of `arXiv:2110.15175`

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Abstract

Problem 4 of Prykarpatsky–Pukach–Vovk–Gregus asks whether certain finite-dimensional bump functions on ℓ_∞^n can have enough height that $(1 + \varepsilon_n)^n \rightarrow \infty$, under the simultaneous constraints $\omega(f; t) \leq t$ and $\omega_2(f; t) \leq t^2$. We give an explicit construction. In fact, for every $n \geq 2$ there is an admissible f_n with

$$\|f_n\|_\infty \geq \frac{1}{8(6 + 8 \log n)}.$$

Consequently $(1 + \varepsilon_n)^n \rightarrow \infty$.

1 Source problem

The source is Y. A. Prykarpatsky, P. Ya. Pukach, M. I. Vovk, and M. Gregus, *Some remarks on smooth mappings of Hilbert and Banach spaces and their local convexity property*, arXiv:2110.15175v2, also published in *Axioms* 13(4), 227 (2024) [1]. The relevant problem appears on page 11 of the arXiv/PDF version as Problem 4; in the TeX source it is labelled `prob2`.

for each $x \in U$ and $h \in X$ with $[x - h, x + h] \subset U$. The Gâteaux differentiability of the function $f : U \rightarrow Y$ implies the differentiability of the function

$$g : [0, 1] \rightarrow Y, \quad g(t) = f(x + th) - 2f(x) + f(x - th).$$

Moreover,

$$g'(t) = f'_{x+th}(h) - f'_{x-th}(h)$$

and hence

$$\|g'(t)\| = \|f'_{x+th} - f'_{x-th}\| \cdot \|h\| \leq L\|2th\| \cdot \|h\| = 2Lt\|h\|^2.$$

Then

$$\begin{aligned} \|f(x + h) - 2f(x) + f(x - h)\| &= \|g(1) - g(0)\| = \\ &= \left\| \int_0^1 g'(t) dt \right\| \leq \int_0^1 \|g'(t)\| dt = 2L \|h\|^2 \int_0^1 t dt = L \|h\|^2. \end{aligned}$$

□

This proposition combined with Theorem 2 implies

Corollary 1. Let X be a Banach space and Y be a Banach space with modulus of convexity of power 2. A homeomorphism $f : U \rightarrow V$ between two open subsets $U \subset X$, $V \subset Y$ of a Banach space X is locally convex if

- the function $f : U \rightarrow V$ is Gâteaux differentiable at each point of U ;
- the derivative map $f' : U \rightarrow L(X, Y)$ is locally Lipschitz;
- the function $f^{-1} : V \rightarrow U$ is locally Lipschitz.

The statement of Theorem 2 is eventually only sufficient. We also at present do not know if the requirement on the convexity modulus of Y is essential in Theorem 2 and Corollary 1.

Remark. Similar to the Hilbert space case, we need to mention here that the integral expressions, considered above with respect to the parameter $t \in [0, 1]$, are well defined in the Banach space Y , as the related mapping $f'_{x(t)}(x - x_0) : [0, 1] \rightarrow Y$, being continuous and of bounded variation, is a priori Riemann-Birkhoff type integrable [8,9,11,13,14,27,32].

Problem 3. Assume that X is a Banach space such that any locally second order Lipschitz homeomorphism $f : X \rightarrow X$ with locally Lipschitz inverse $f^{-1} : X \rightarrow X$ is locally convex. Is X (super)reflexive?

Problem 4. For every $n \in \mathbb{N}$ let F_n be the set of all functions $f_n : l_\infty(n) \rightarrow [0, 1]$ on the n -dimensional Banach space $l_\infty(n) = (\mathbb{R}^n, \|\cdot\|_\infty)$ such that

- $f_n^{-1}(0, 1] \subset (-1, 1)^n$;
- $\omega(f; t) \leq t$ for all $t \geq 0$;
- $\omega_2(f; t) \leq t^2$ for all $t \geq 0$.

Let also $\varepsilon_n = \sup\{\|f_n\|_\infty : f_n \in F_n\}$. Is $\lim_{n \rightarrow \infty} (1 + \varepsilon_n)^n = \infty$?

If the Problem 4 has an affirmative answer, then Problem 3 has negative answer. Namely, on the reflexive Banach space $X = (\sum_{n=1}^\infty l_\infty(n))_{l_2}$ there is a homeomorphism $f : X \rightarrow X$ which is not locally convex but $f : X \rightarrow X$ is second order Lipschitz and $f^{-1} : X \rightarrow X$ is locally Lipschitz.

3. Conclusion

The paper analyses of smooth nonlinear mappings for Hilbert and Banach spaces that carry small balls to convex sets, provided that the radius of the balls is sufficiently small. The main goal is to establish new and mild sufficient conditions for a nonlinear mapping to

Transcribed, the problem defines F_n to be the set of functions $f_n : l_\infty(n) \rightarrow [0, 1]$, where

$\ell_\infty(n) = (\mathbb{R}^n, \|\cdot\|_\infty)$, such that

$$f_n^{-1}(0, 1] \subset (-1, 1)^n, \quad \omega(f; t) \leq t, \quad \omega_2(f; t) \leq t^2 \quad (t \geq 0).$$

With

$$\varepsilon_n = \sup\{\|f_n\|_\infty : f_n \in F_n\},$$

the question is whether

$$\lim_{n \rightarrow \infty} (1 + \varepsilon_n)^n = \infty.$$

Here $\omega = \omega_1$ is the ordinary modulus of continuity and ω_2 is the second-difference modulus recalled in the source:

$$\omega_2(f; t) = \sup_{\|h\|_\infty \leq t} |f(x+h) - 2f(x) + f(x-h)|.$$

The domain here is all of $\mathbb{R}^n$, so no segment restriction is needed.

2 Proof intuition

The obstruction is the flat faces of the cube for ℓ_∞^n : a bump has to vanish before it reaches any coordinate wall, while the first and second finite differences are measured in the sup norm. A product bump loses height exponentially in n . Instead we use a soft minimum of the coordinate wall distances $1 - x_i^2$. The logarithmic loss in the soft minimum is only $(\log n)/\beta$. Taking $\beta = 2 \log n$ leaves height $1/2$ at the origin, while the Hessian cost is only $O(\log n)$. After scaling by that Hessian constant we retain height comparable to $1/\log n$, enough for $(1 + \varepsilon_n)^n$ to diverge.

3 Construction and estimates

For $n = 1$ set $f_1 = 0$. Fix $n \geq 2$ and put

$$\beta = 2 \log n.$$

For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ define

$$s_i(x) = 1 - x_i^2, \quad M(x) = -\frac{1}{\beta} \log \sum_{i=1}^n e^{-\beta s_i(x)} = -\frac{1}{\beta} \log \sum_{i=1}^n e^{\beta(x_i^2 - 1)}.$$

Let

$$\psi(u) = \frac{1}{2}(u_+)^2, \quad G(x) = \psi(M(x)), \quad f_n(x) = \frac{G(x)}{6 + 4\beta}.$$

Lemma 1 (support and height). *The function f_n maps $\mathbb{R}^n$ into $[0, 1]$, satisfies $f_n^{-1}(0, 1] \subset (-1, 1)^n$, and*

$$\|f_n\|_\infty \geq \frac{1}{8(6 + 4\beta)} = \frac{1}{8(6 + 8 \log n)}.$$

Proof. Since the sum in the definition of M dominates each one of its terms,

$$M(x) \leq s_i(x) \quad (1 \leq i \leq n).$$

Thus $M(x) > 0$ implies $s_i(x) > 0$ for every i , i.e. $|x_i| < 1$ for every i . This proves the support condition for G and hence for f_n .

Also, $s_i(x) \leq 1$ for all i , so

$$\sum_{i=1}^n e^{-\beta s_i(x)} \geq n e^{-\beta}.$$

Therefore

$$M(x) \leq 1 - \frac{\log n}{\beta} = \frac{1}{2}.$$

Hence $0 \leq G \leq \frac{1}{2}(1/2)^2 = 1/8$, and $0 \leq f_n \leq 1$ is immediate. At $x = 0$ one has

$$M(0) = -\frac{1}{\beta} \log(n e^{-\beta}) = 1 - \frac{\log n}{\beta} = \frac{1}{2},$$

so $G(0) = 1/8$ and the displayed lower bound follows. $\square$

Lemma 2 (differential bounds). *For every $x, h \in \mathbb{R}^n$,*

$$|DG(x)h| \leq \|h\|_\infty$$

where G is differentiable. Moreover, for every line $u(t) = G(x + th)$ the derivative u' is absolutely continuous and

$$|u''(t)| \leq (6 + 4\beta)\|h\|_\infty^2$$

for almost every $t \in \mathbb{R}$.

Proof. Set

$$p_i(x) = \frac{e^{-\beta s_i(x)}}{\sum_{j=1}^n e^{-\beta s_j(x)}}.$$

These are nonnegative weights with $\sum_i p_i = 1$. The first derivative of the log-sum-exp soft minimum is

$$DM(x)h = \sum_{i=1}^n p_i(x) Ds_i(x)h = -2 \sum_{i=1}^n p_i(x) x_i h_i.$$

At points where $M(x) > 0$, the preceding support argument gives $|x_i| < 1$ for all i , and hence

$$|DM(x)h| \leq 2\|h\|_\infty.$$

At points where $M(x) \leq 0$, $\psi'(M(x)) = 0$. Since $0 \leq M_+(x) \leq 1/2$, the chain rule gives

$$|DG(x)h| = |\psi'(M(x))DM(x)h| \leq \frac{1}{2} \cdot 2\|h\|_\infty = \|h\|_\infty.$$

For the second estimate, first compute the second directional derivative of M . With $a_i = Ds_i(x)h = -2x_i h_i$ and $D^2 s_i(x)[h, h] = -2h_i^2$, the standard log-sum-exp formula gives

$$D^2 M(x)[h, h] = \sum_i p_i D^2 s_i(x)[h, h] - \beta \left(\sum_i p_i a_i^2 - \left(\sum_i p_i a_i \right)^2 \right).$$

At points with $M(x) > 0$,

$$\left| \sum_i p_i D^2 s_i(x)[h, h] \right| \leq 2\|h\|_\infty^2,$$

and, since $|a_i| \leq 2\|h\|_\infty$,

$$0 \leq \sum_i p_i a_i^2 - \left(\sum_i p_i a_i \right)^2 \leq 4\|h\|_\infty^2.$$

Therefore

$$|D^2 M(x)[h, h]| \leq (2 + 4\beta)\|h\|_\infty^2 \quad (M(x) > 0).$$

The function $\psi(u) = \frac{1}{2}(u_+)^2$ is C^1 and has Lipschitz derivative; its second derivative exists a.e. and satisfies $0 \leq \psi'' \leq 1$. Along any line $x + th$, the composition $u(t) = G(x + th) = \psi(M(x + th))$ is $C^{1,1}$, hence u' is absolutely continuous, and for a.e. t with $M(x + th) > 0$,

$$\begin{aligned} |u''(t)| &\leq |DM(x + th)h|^2 + M(x + th)|D^2 M(x + th)[h, h]| \\ &\leq 4\|h\|_\infty^2 + \frac{1}{2}(2 + 4\beta)\|h\|_\infty^2 \\ &\leq (6 + 4\beta)\|h\|_\infty^2. \end{aligned}$$

Where $M(x + th) < 0$, u is locally zero, and the same bound is trivial. The a.e. estimate also holds across the zero set because ψ' is continuous and Lipschitz, so there is no jump contribution in u' . $\square$

Theorem 1. *Problem 4 has an affirmative answer. More quantitatively,*

$$\varepsilon_n \geq \frac{1}{8(6 + 8 \log n)} \quad (n \geq 2),$$

and hence

$$\lim_{n \rightarrow \infty} (1 + \varepsilon_n)^n = \infty.$$

Proof. The support and range conditions were proved above. It remains to check the two moduli. If $x, y \in \mathbb{R}^n$, apply the first derivative bound to the segment $x + t(y - x)$:

$$|G(y) - G(x)| \leq \|y - x\|_\infty.$$

Since $0 < (6 + 4\beta)^{-1} \leq 1$, the same estimate holds for f_n . Thus $\omega(f_n; t) \leq t$ for all $t \geq 0$.

For the second modulus, fix $x, h \in \mathbb{R}^n$ and put $u(t) = G(x + th)$. The one-dimensional identity

$$u(1) - 2u(0) + u(-1) = \int_{-1}^1 (1 - |t|)u''(t) dt$$

holds for functions with absolutely continuous derivative. Therefore

$$|G(x + h) - 2G(x) + G(x - h)| \leq (6 + 4\beta)\|h\|_\infty^2.$$

After multiplying by $(6 + 4\beta)^{-1}$, we get

$$|f_n(x + h) - 2f_n(x) + f_n(x - h)| \leq \|h\|_\infty^2.$$

Taking the supremum over $\|h\|_\infty \leq t$ gives $\omega_2(f_n; t) \leq t^2$ for all $t \geq 0$. Hence $f_n \in F_n$, and the height bound from the first lemma gives the displayed estimate for ε_n .

Finally,

$$\log(1 + \varepsilon_n)^n = n \log(1 + \varepsilon_n) \geq n \log \left(1 + \frac{1}{8(6 + 8 \log n)} \right) \rightarrow \infty,$$

because the last expression grows on the order of $n/\log n$. This proves the claim. $\square$

4 Verification notes

The proof is analytic and does not rely on computation. The included verifier script in `code/` performs deterministic random sampling for $n = 2, 3, 5, 10, 25$ to check three elementary failure modes: positivity outside the open cube, violation of the Lipschitz scaling in sampled pairs, and violation of the second-difference scaling in sampled triples. Its sampled ratios stay below the theoretical bound and it reports no support violations.

5 Novelty and literature check

Bounded searches were performed on 2026-07-19 before promotion. The local run indexes were searched for 2110.15175, the exact title, local convexity phrases, and the displayed ε_n problem. Web searches used the exact arXiv id, exact title, DOI 10.3390/axioms13040227, Problem 4, `omega.2(f;t)`, `l_infty(n)`, and the limit expression $(1 + \varepsilon_n)^n$. The hits identified the arXiv record, the published Axioms/MDPI article, and mirrors or citation pages. No separate paper or theorem claiming an answer to Problem 4, or a lower bound of the form $\varepsilon_n \gtrsim 1/\log n$, was found in this bounded search. This is not a proof of novelty, but no duplicate answer was detected.

6 Human-review recommendation

The main point for review is the $C^{1,1}$ line-restriction argument for $\psi(u) = \frac{1}{2}(u_+)^2$. Once that standard a.e. second derivative estimate is accepted, the construction gives the required admissible functions and the divergence follows immediately. The result also triggers the source paper's stated implication that an affirmative answer to Problem 4 gives a negative answer to its Problem 3, subject to the source paper's argument for that implication.

References

- [1] Y. A. Prykarpatsky, P. Ya. Pukach, M. I. Vovk, and M. Gregus, *Some remarks on smooth mappings of Hilbert and Banach spaces and their local convexity property*, arXiv:2110.15175v2; Axioms **13**(4), 227 (2024), doi:10.3390/axioms13040227.